Notice, question, predict

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use observations to make a prediction that can be tested.

SUCCESS CRITERIA - I can...

1. Separate stated evidence from unsupported claims.
2. Extend a position pattern and state an assumption.
3. Name an observation that could test the prediction.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “prediction” to explain your model or result. Sentence starters are optional.

evidence: observations or measurements that support an idea

prediction: what a model says may happen next

model: a drawing, explanation or equation for a physical situation

01 Read a scale

A ruler shows a toy car at 2 cm, then at 7 cm. **Draw** both positions and the change. How far apart are the marks?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta x = x_f - x_i$; $x_{next} = x_{now} + \Delta x$.
For a repeating position pattern only. i = initial; f = final. Check the pattern against evidence.

Worked reasoning: A cart is at 2, 5 and 8 m at 0, 1 and 2 s. Equal jumps of 3 m suggest 11 m at 3 s, if the pattern continues.

Your turn: A motion sensor recorded the first three positions in the table. Add the predicted position at 3 s. **Draw** the four positions on a line and distinguish the prediction from the measurements.

Sentence starter: I predict ... . My model supports this because ... .

Time (s)	Position (m)
0	2
1	5
2	8
3	Predict: ____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Extract the evidence

A fictional space report says: "Repeated images show an object approaching Earth. Its path and size remain uncertain." Identify two statements the report supports and one conclusion it does not support.
Explain the difference between an observation and a forecast.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Fit a simple model

A motion sensor tracks a cart at one-second intervals.
Use the table to write a position rule, x(t), and predict x at 5 s.
Explain the assumption that lets you extend the pattern.

Sentence starter: I predict … . My model supports this because … .

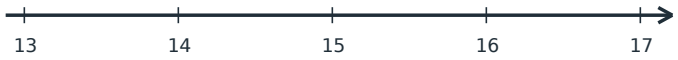
t (s)	x (m)
0	1
1	4
2	7
3	10

Marzano check - circle: ① ② ③

05 Challenge the prediction

At 5 s, the sensor measures 14.0 ± 0.2 m.
Mark this interval and your prediction from question 4 on the number line. Do they agree?
Give two possible reasons for any disagreement.

Sentence starter: I predict … . My model supports this because … .



Position (m); right is positive

Marzano check - circle: ① ② ③

06 Design the observation

A telescope team can make two more observations of the fictional asteroid.
Describe what to measure and when.
Explain how the pair of measurements would improve a prediction of its future position.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

07 Compare claims

Claim A: “A straight-line model predicts where it may be.” Claim B: “The asteroid will hit Earth.” Explain what additional evidence and physics separate these claims.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

08 Choose useful evidence

Two groups time the same toy car over the same marked track. One records three times; the other writes "pretty fast." Which record would let someone check the claim? **Explain**.

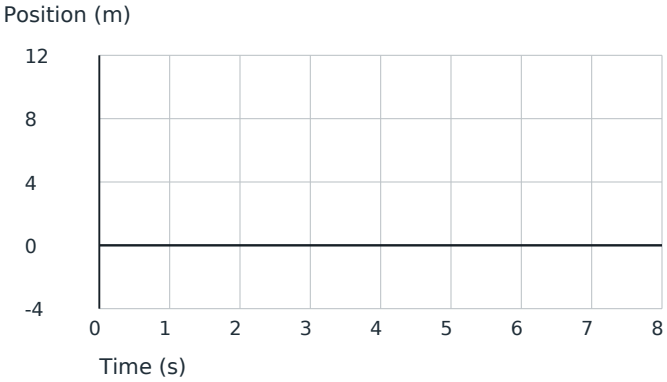
Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

09 Exit check

A robot's positions are 12, 8 and 4 m at 0, 2 and 4 s. Plot them, determine the velocity, and predict its position at 7 s.
State an assumption and a later measurement that could test it.

Sentence starter: I predict … . My model supports this because … .

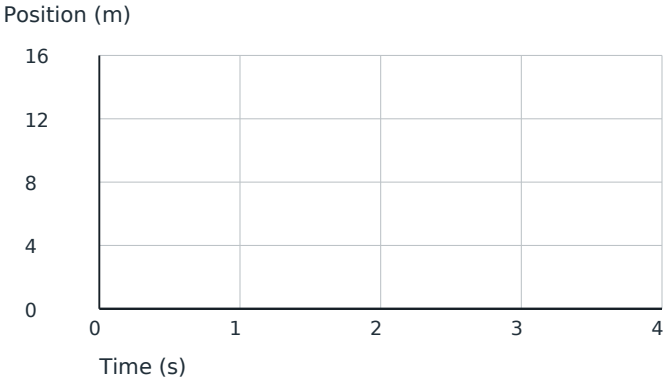


Marzano check - circle: ① ② ③

10 STRETCH: test the model

Return to the worked example in problem 2. Construct two position rules that match all three observations in the worked case but disagree at 4 s.
Explain why the data do not guarantee either extrapolation.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: ① ② ③